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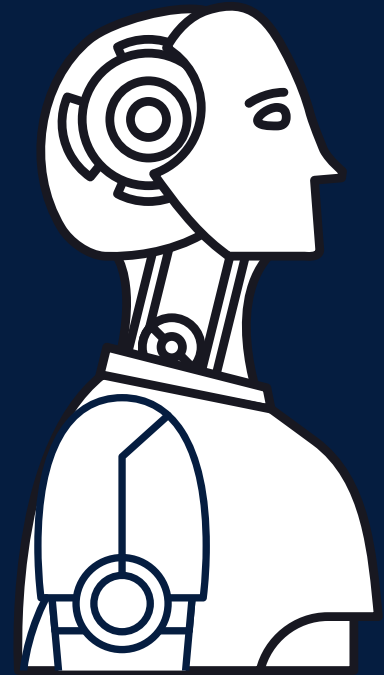


SAE21

CONSTRUIRE UN RÉSEAU INFORMATIQUE



ENSEIGNANT : MR. OUAMRI



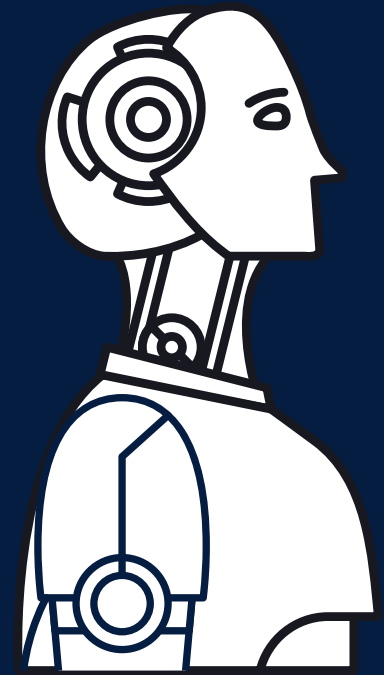
PRÉSENTÉ PAR:
BRINGUIER TRISTAN
AHMAD IRSA
CHADLI JAWED





INTRODUCTION

La mise en place des réseaux informatiques afin d'assurer la gestion, la sécurité et l'optimisation des infrastructures et de garantir une connectivité fiable et une protection contre les menaces.



INTRODUCTION

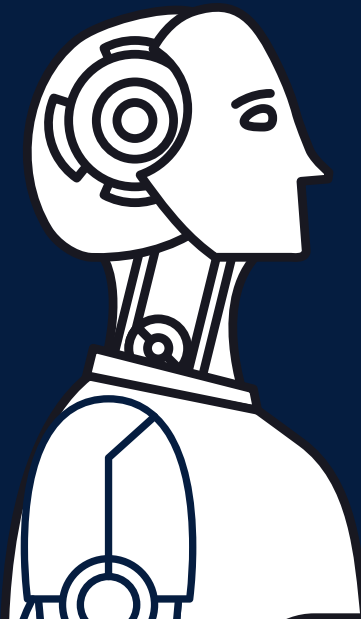


OBJECTIFS CLÉS :

- Mettre en pratique les différents protocoles réseaux vus durant l'année
- Prendre en main des équipements de la marque Cisco

MATÉRIEL UTILISÉ :

- Switchs CISCO Catalyst 2960-C Series LL
- Routeurs CISCO 4200 Series
- Ordinateurs sur Debian 11 (bullseye)

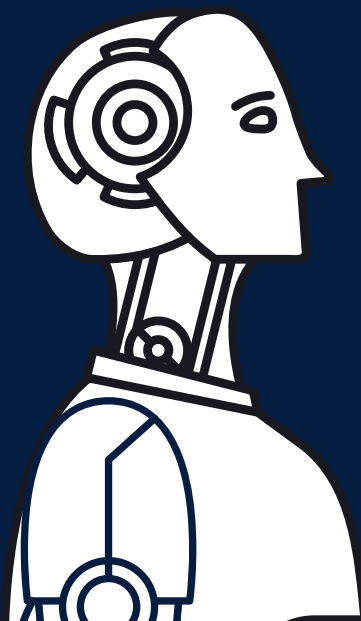


SOMMAIRE



NOTIONS ABORDÉES :

- L'adressage IP
- L'ARP
- L'ICMP
- Les VLANs
- Les routes statiques
- Le protocole RIPv2
- Le protocole OSPF
- Le DHCP (Server/Relay/Client)
- NAT/PAT Static
- NAT/PAT Dynamic

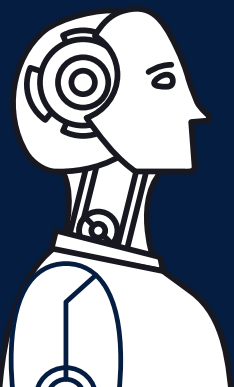




L'ADRESSAGE IP



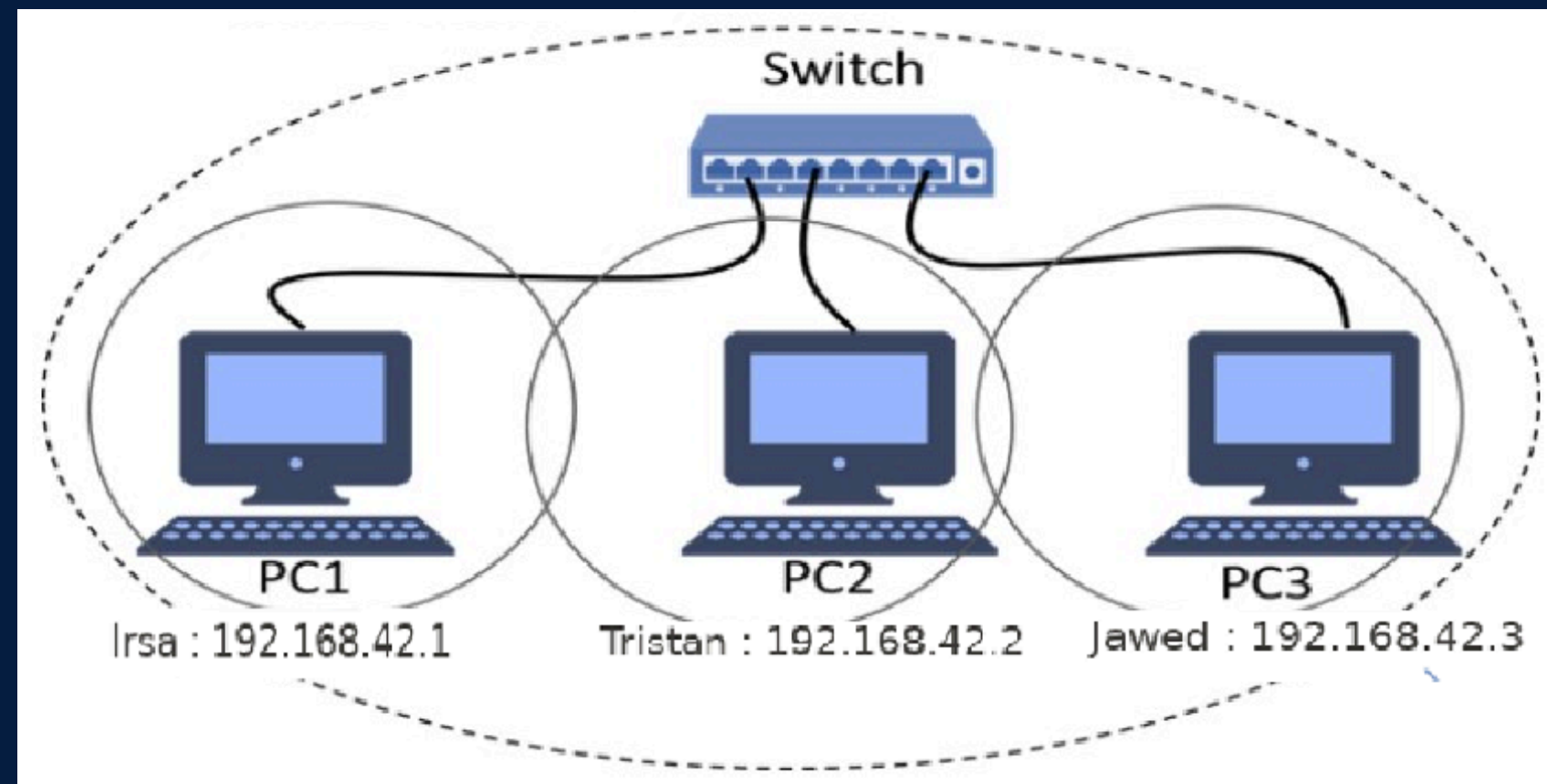
Etape cruciale pour toute construction de réseau informatique, permettant l'identification et la communication entre toute périphérique se trouvant sur le réseau.



L'ADRESSAGE IP

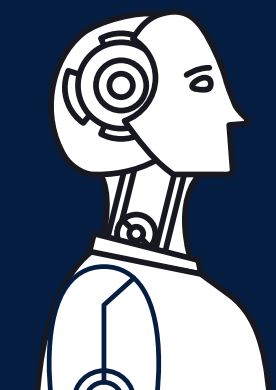
Topologie en étoile avec trois machines connectées entre elles avec le Switch Cisco.

Adresses de classe C 192.168.42.x port eth0





L'ARP





ARP

Address Resolution Protocol

- Association des adresses IP et MAC
- Table d'ARP
- Cache ARP

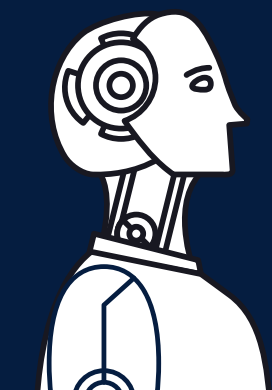
2	0.216750957	SpeedDra_e7:10:b6	Broadcast	ARP	42 Who has 192.168.42.3? Tell 192.168.42.2
3	0.217124055	SpeedDra_20:e6:03	SpeedDra_e7:10:b6	ARP	60 192.168.42.3 is at 00:13:3b:20:e6:03
4	0.217130375	192.168.42.2	192.168.42.3	ICMP	98 Echo (ping) request id=0x5d70, seq=1/256, ttl=64 (reply in 5)
5	0.217398936	192.168.42.3	192.168.42.2	ICMP	98 Echo (ping) reply id=0x5d70, seq=1/256, ttl=64 (request in 4)

Adresse IP : 12.34.56.78

Adresse MAC : A1:B2:C3:D4:EF:67



L'ICMP





ICMP

Internet Control Message Protocol

- Diagnostiquer les réseaux
- Etudier les chemins des réseaux
- Calculer la latence

4	0.217130375	192.168.42.2	192.168.42.3	ICMP	98 Echo (ping) request	id=0x5d70, seq=1/256, ttl=64 (reply in 5)
5	0.217398936	192.168.42.3	192.168.42.2	ICMP	98 Echo (ping) reply	id=0x5d70, seq=1/256, ttl=64 (request in 4)
6	1.242110319	192.168.42.2	192.168.42.3	ICMP	98 Echo (ping) request	id=0x5d70, seq=2/512, ttl=64 (reply in 7)
7	1.242654774	192.168.42.3	192.168.42.2	ICMP	98 Echo (ping) reply	id=0x5d70, seq=2/512, ttl=64 (request in 6)

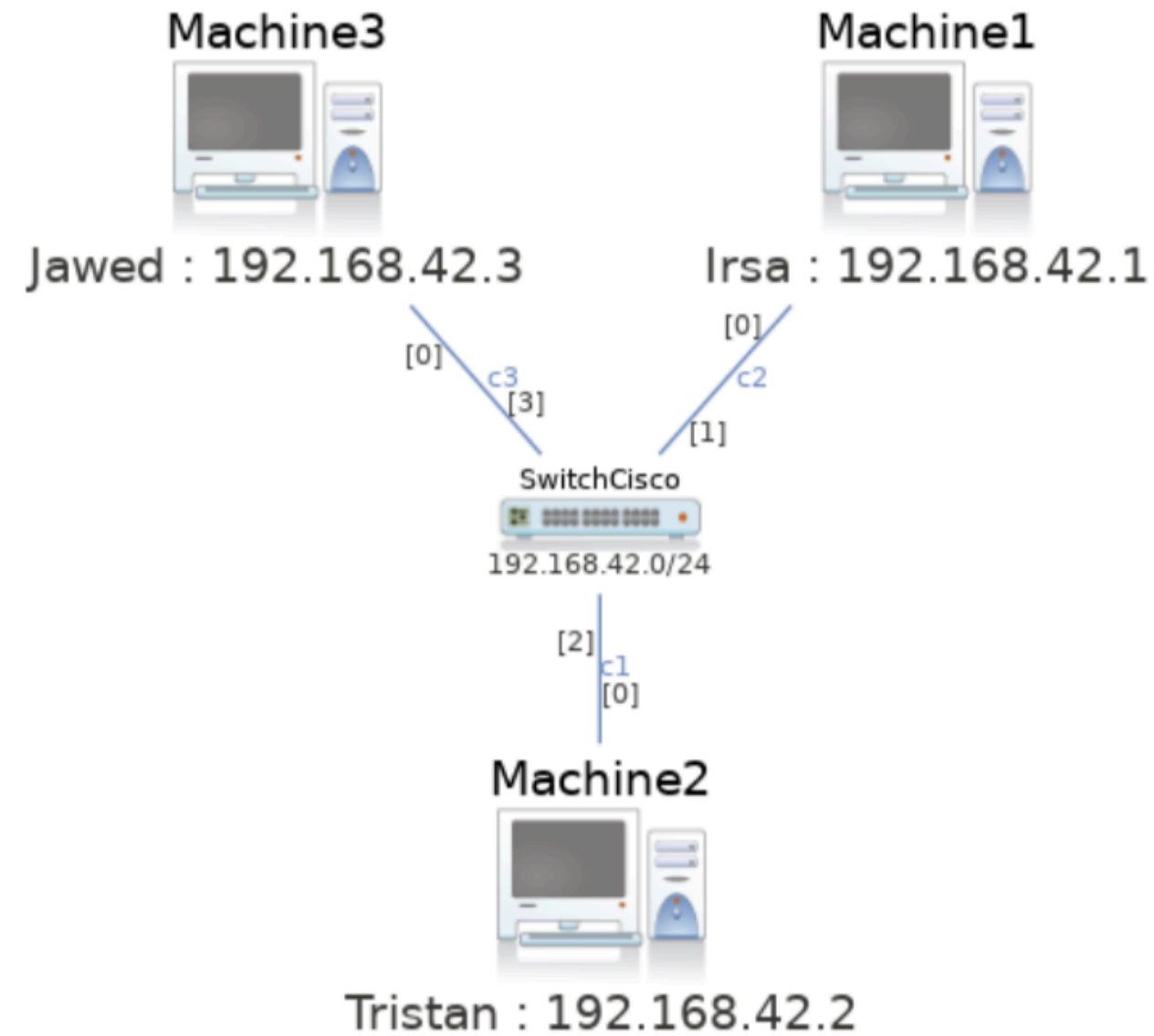
Echo Reply (0)

Destination Unreachable (3)

Redirect (5)

Time Exceeded (11)





```
root@p20208:~# ping 192.168.42.254
PING 192.168.42.254 (192.168.42.254) 56(84) bytes of data.
64 bytes from 192.168.42.254: icmp_seq=1 ttl=255 time=6.47 ms
64 bytes from 192.168.42.254: icmp_seq=2 ttl=255 time=0.402 ms
```

2	0.216750957	SpeedDra_e7:10:b6	Broadcast	ARP	42 Who has 192.168.42.3? Tell 192.168.42.2
3	0.217124055	SpeedDra_20:e6:03	SpeedDra_e7:10:b6	ARP	60 192.168.42.3 is at 00:13:3b:20:e6:03
4	0.217130375	192.168.42.2	192.168.42.3	ICMP	98 Echo (ping) request id=0x5d70, seq=1/256, ttl=64 (reply in 5)
5	0.217398936	192.168.42.3	192.168.42.2	ICMP	98 Echo (ping) reply id=0x5d70, seq=1/256, ttl=64 (request in 4)
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7	1.242654774	192.168.42.3	192.168.42.2	ICMP	98 Echo (ping) reply id=0x5d70, seq=2/512, ttl=64 (request in 6)

▼ Address Resolution Protocol (request)

Hardware type: Ethernet (1)

Protocol type: IPv4 (0x0800)

Hardware size: 6

Protocol size: 4

Opcode: request (1)

Sender MAC address: SpeedDra_e7:10:b6 (00:13:3b:e7:10:b6)

Sender IP address: 192.168.42.2

Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)

Target IP address: 192.168.42.3

▼ Internet Control Message Protocol

Type: 8 (Echo (ping) request)

Code: 0

Checksum: 0x6c85 [correct]

[Checksum Status: Good]

Identifier (BE): 23920 (0x5d70)

Identifier (LE): 28765 (0x705d)

Sequence Number (BE): 1 (0x0001)

Sequence Number (LE): 256 (0x0100)

[Response frame: 5]

Timestamp from icmp data: May 14, 2024 09:20:39.000000000 CEST

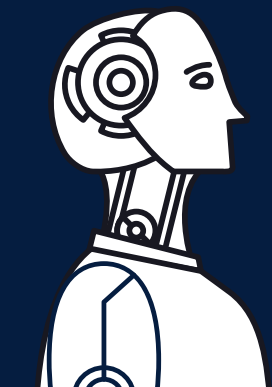
[Timestamp from icmp data (relative): 0.508138216 seconds]

▼ Data (48 bytes)

Data: 5dbf07000000000000101112131415161718191a1b1c1d1e1f202122232425262728292a2b...

[Length: 48]

LE VLAN

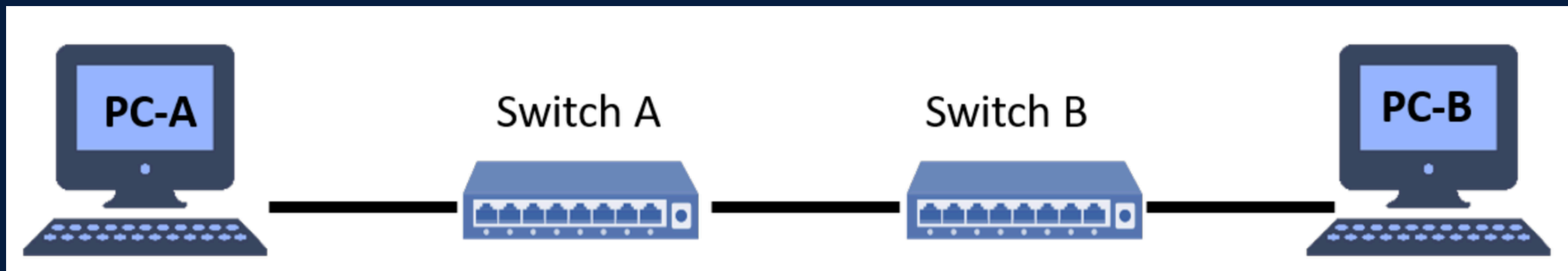


VLAN

Virtual Local Area Network

4096

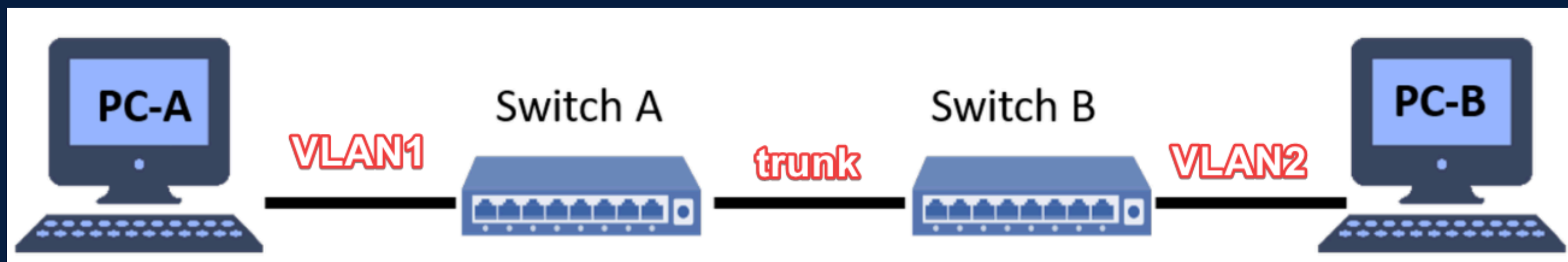
- Isolation des réseaux locaux
- Sécurisation des réseaux
- Possibilités de routage avancées



```
S1(config)# vlan 10
S1 (config-vlan) # name Operations
S1(config-vlan)# vlan 20
S1 (config-vlan) # nom Parking_Lot
S1(config-vlan)# vlan 99
S1(config-vlan)# name Management
S1 (config-vlan) # vlan 1000
S1(config-vlan)# name Native
S1(config-vlan)# end
```

```
S1(config)# interface f0/6
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 10
```

```
S1(config)# interface f0/1
S1(config-if)# switchport mode trunk
```



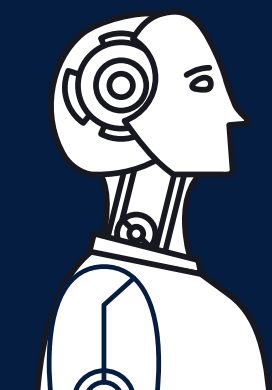
Réseau étanche !!!



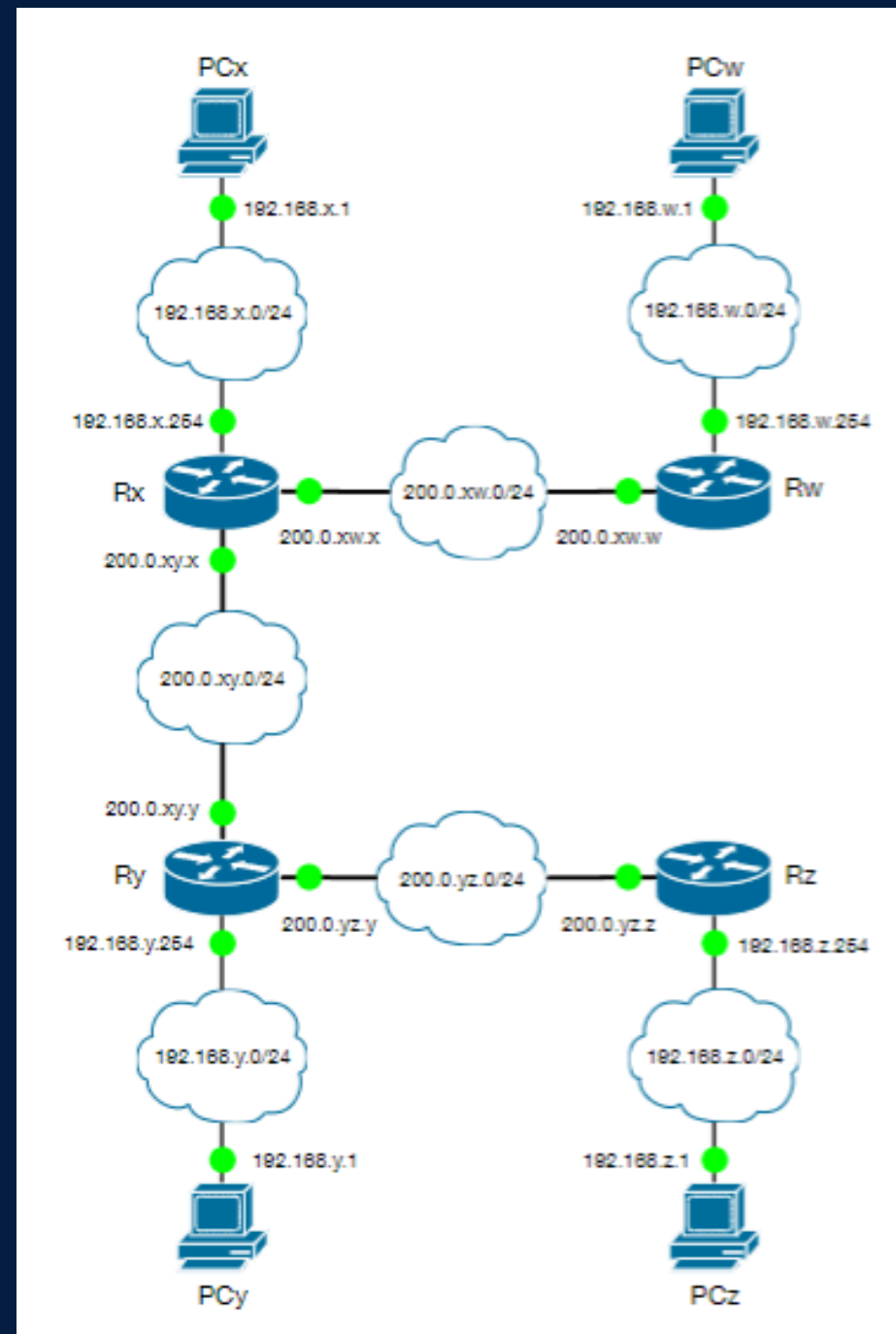
```
▸ Ethernet II, Src: c2:06:1b:44:00:00 (c2:06:1b:44:00:00), Dst: c2:02:18:43:00:00 (c2:02:18:43:00:00)
▸ 802.1Q Virtual LAN, PRI: 0, CFI: 0, ID: 2
    000. .... .... = Priority: Best Effort (default) (0)
    ...0 .... .... = CFI: Canonical (0)
    .... 0000 0000 0010 = ID: 2
    Type: IPv4 (0x0800)
▸ Internet Protocol Version 4, Src: 10.1.3.4, Dst: 10.1.2.1
▸ Internet Control Message Protocol
```



LE ROUTAGE STATIQUE



ROUTAGE STATIQUE



Quelques commandes :

```
R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ip route 192.168.32.0 255.255.255.0 200.0.42.253
R2(config)#exit
```

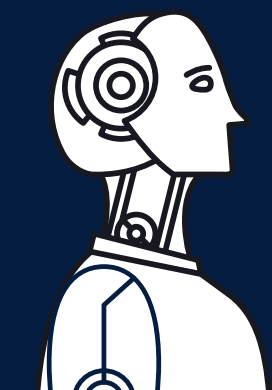
```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m - CMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       H - NHRP, G - NHRP registered, g - NHRP registration summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from P/R
       & - replicated local route overrides by connected
```

Gateway of last resort is not set

```
S    192.168.32.0/24 [1/0] via 200.0.42.253
C    192.168.42.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.42.0/24 is directly connected, GigabitEthernet0/0/0
L    192.168.42.254/32 is directly connected, GigabitEthernet0/0/0
C    200.0.42.0/24 is variably subnetted, 2 subnets, 2 masks
C    200.0.42.0/24 is directly connected, GigabitEthernet0/0/1
L    200.0.42.254/32 is directly connected, GigabitEthernet0/0/1
R2#
```



RIP



RIP

Routing Information Protocol

- Routage dynamique
- Quelques commandes :

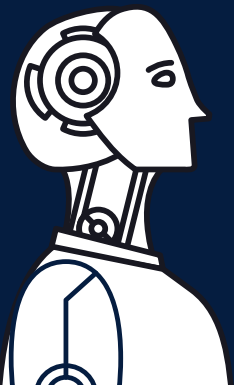
```
Rx(config)#router rip
Rx(config)#version 2
Rx(config-router)#network 192.168.x.0
Rx(config-router)#network 200.0.xy.0
```

- Analyse d'un paquet :

```
4 Ethernet II, Src: 74:11:b2:f3:39:51 (74:11:b2:f3:39:51), Dst: IPv4mcast_09 (01:00:5e:00:00:09)
  1 Internet Protocol Version 4, Src: 200.0.2.253, Dst: 224.0.0.9
  2 User Datagram Protocol, Src Port: 520, Dst Port: 520
  3 Routing Information Protocol
    Command: Response (2)
    Version: RIPv2 (2)
    - IP Address: 192.168.42.0, Metric: 2
      Address Family: IP (2)
      Route Tag: 0
      IP Address: 192.168.42.0
      Netmask: 255.255.255.0
      Next Hop: 0.0.0.0
      Metric: 2
    - IP Address: 200.0.3.0, Metric: 1
      Address Family: IP (2)
      Route Tag: 0
      IP Address: 200.0.3.0
      Netmask: 255.255.255.0
      Next Hop: 0.0.0.0
      Metric: 1
```



OSPF





OSPF

Open Shortest Path First

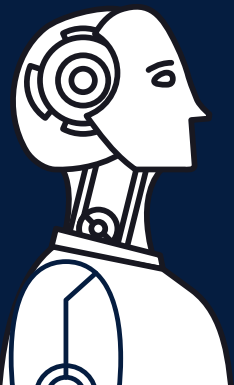
- Routage dynamique
- Quelques commandes :

```
Rx(config)#router ospf 1  
Rx(config-router)#network 192.168.x.0 0.0.0.255 area 0  
Rx(config-router)#network 200.0.xy.0 0.0.0.255 area 0
```





DHCP

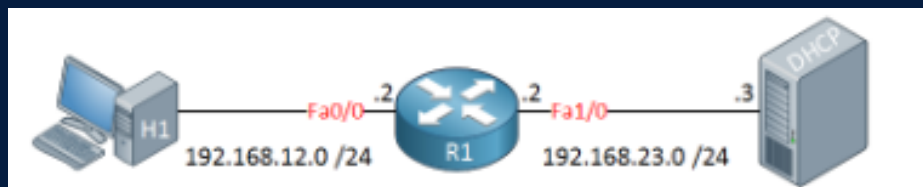


DHCP

Dynamic Host Configuration Protocol

- Attribution d'adresses IP
- Informations sur certaines machines du réseau

```
DHCP(config)#ip dhcp pool PoolOnRx
DHCP(dhcp-config)#default-router 192.168.12.254
DHCP(dhcp-config)#dns-server 208.67.222.222
```



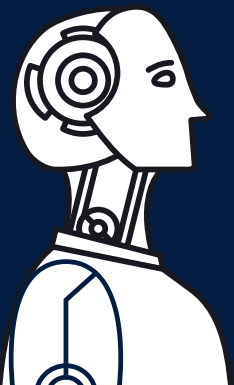
```
DHCP#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration    Type      State      Interf
ce
                Hardware address/
                User name
192.168.12.1    0013.3be7.10b6   May 16 2024 07:17 AM Automatic Active   Gigabi
Ethernet0/0/0
DHCP#
```



NAT/PAT



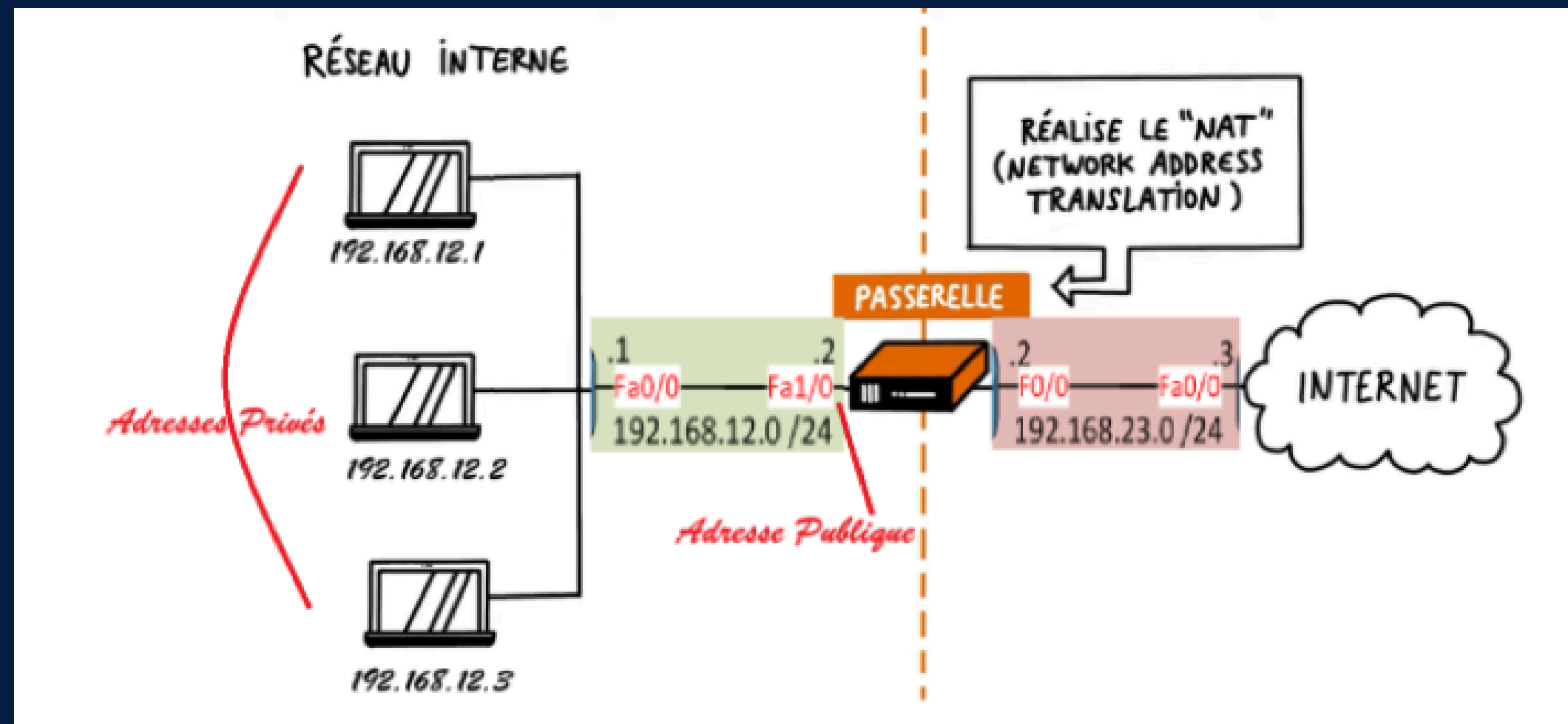
La communication entre les machines de différents réseaux peut se faire en jouant sur le port source et l'adresse IP publique, avec le NAT et le PAT.



NAT STATIQUE

Network Address Translation

› Traduction des adresses internes privées en publiques

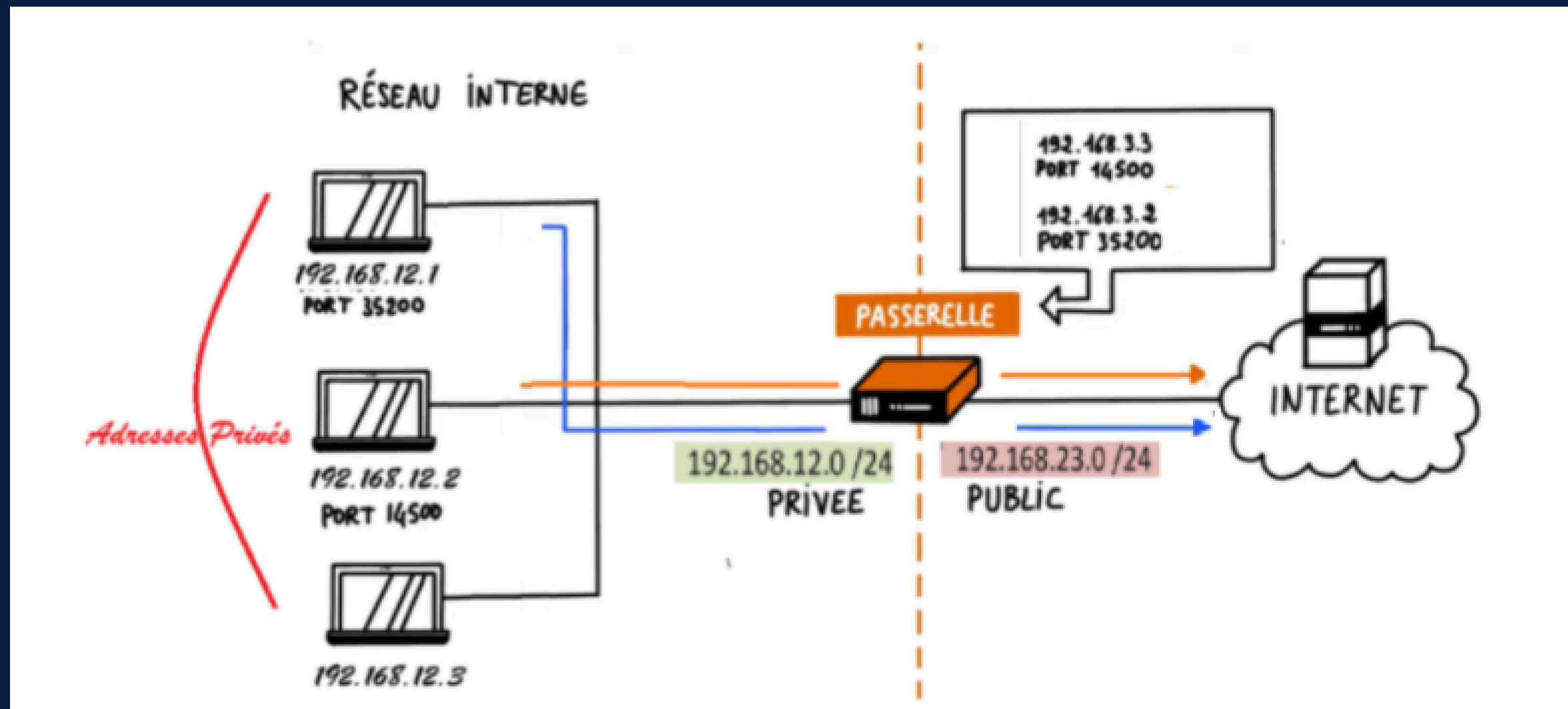


PAT STATIQUE

Port Address Translation

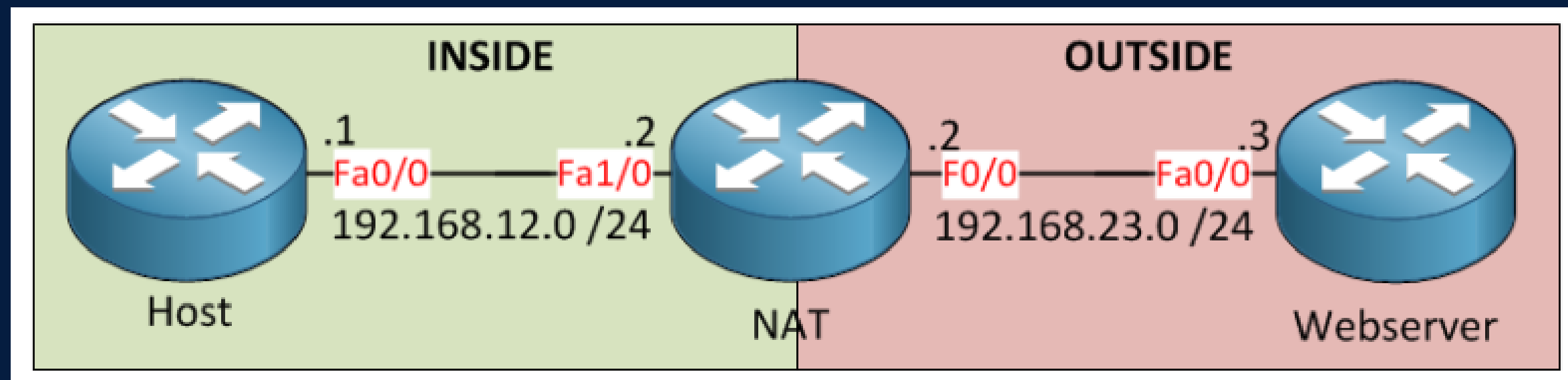
Lorsque plusieurs machines envoient des requêtes en même temps > la passerelle réalise le PAT

- > Attribution un port spécifique à chaque machine faisant une demande de communication
- > Ainsi qu'une adresse IP publique



NAT/PAT STATIQUE

> La Topologie

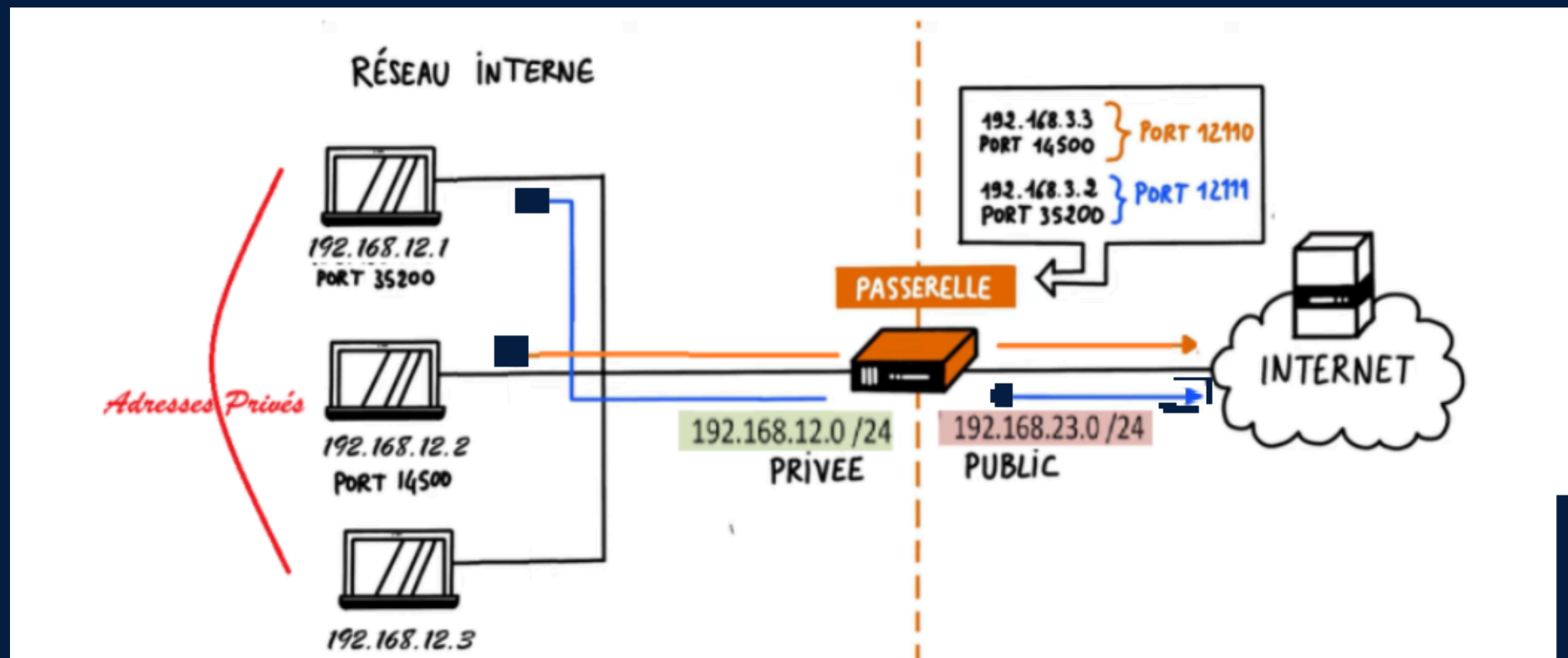


> Configuration sur le routeur

```
NAT(config)# ip nat inside source static 192.168.12.1  
192.168.23.2
```

NAT/PAT DYNAMIQUE

Permet la traduction d'une adresse IP privée à un pool d'adresses IP publiques configurées sur le routeur.

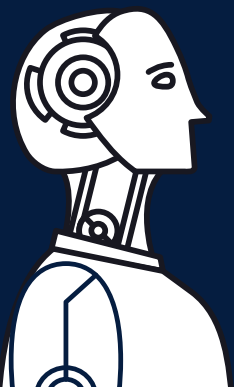


NAT/PAT DYNAMIQUE



› La configuration nécessaire sur la passerelle:

```
NAT(config)#access-list 1 permit 192.168.123.0 0.0.0.255
NAT(config)#ip nat inside source list 1 interface fastEthernet
    1/0 overload
NAT(config)#interface fastEthernet 0/0
NAT(config-if)#ip nat inside
NAT(config-if)#exit
NAT(config)#int fastEthernet 0/1
NAT(config-if)#ip nat outside
NAT(config-if)#exit
```



CONCLUSION



- Matériel Cisco
- Protocoles utilisés en entreprise
- Compétences basique d'administration réseau

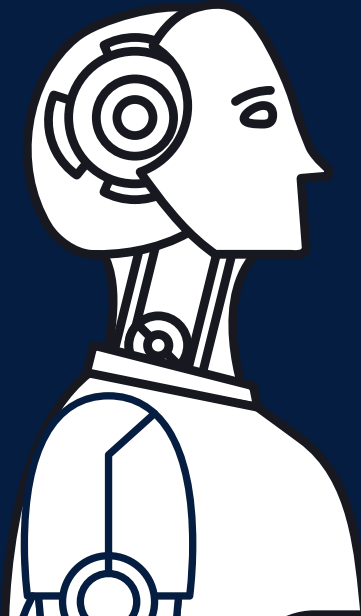
RT1.2 - Administrer les réseaux et l'Internet



RT2.2 - Connecter les entreprises et les usagers



RT3.2 - Créer des outils et applications informatiques pour les R&T





MERCI DE VOTRE ATTENTION.

